



Explore Careers. Learn Stuff. Make Connections.

Zrive Applied Data Science

Fundamentals of Statistical Learning

Guillermo Barquero

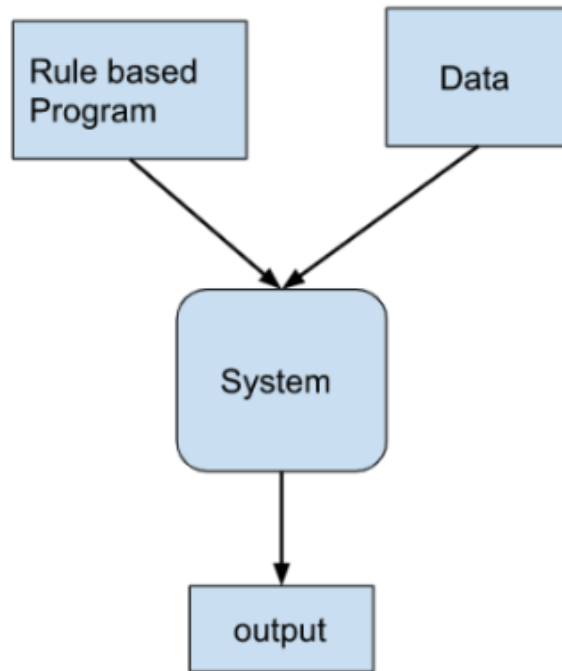
Today's concepts

- 1 What it is to learn?
- 2 Supervised Learning
- 3 Bias-variance trade off
- 4 Linear models & regularisation
- 5 Train-validation-test split
- 6 Metrics

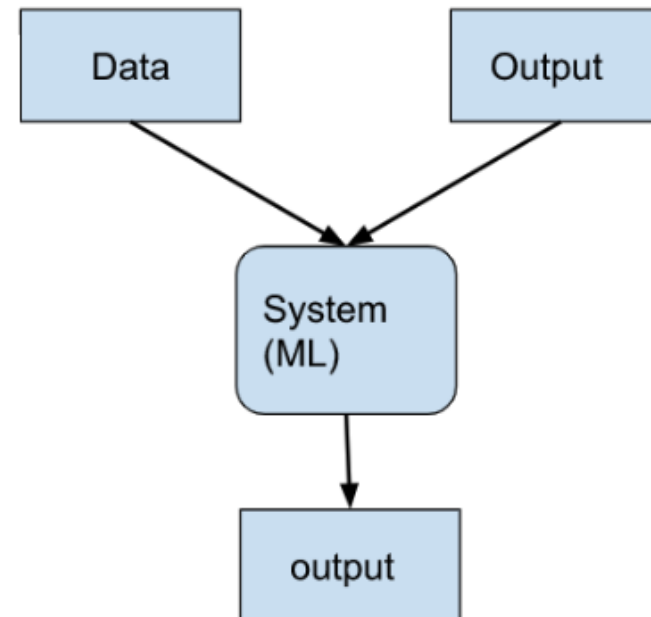
Statistical Learning

1 What is machine learning?

It deals with finding a predictive function based on the data presented. The main idea in machine learning is to build a model that can draw conclusions from data and make predictions.



Rule-based model

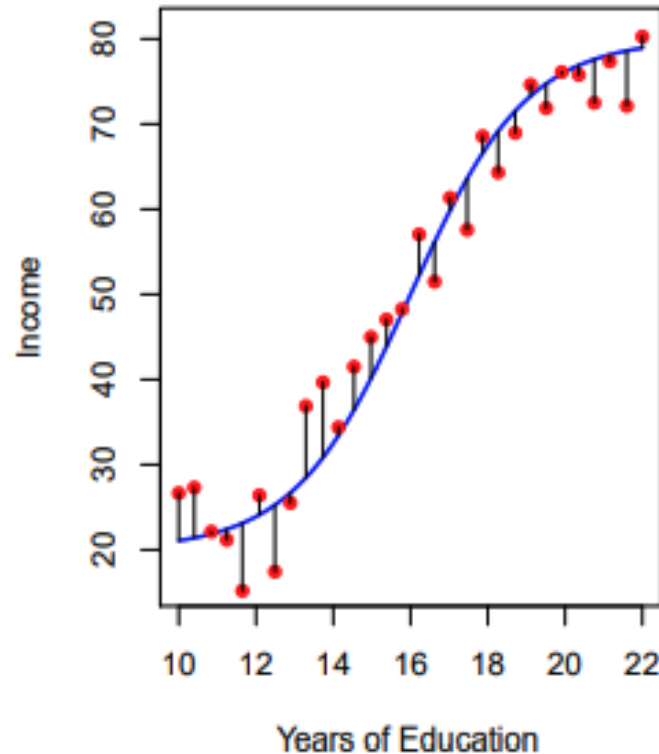
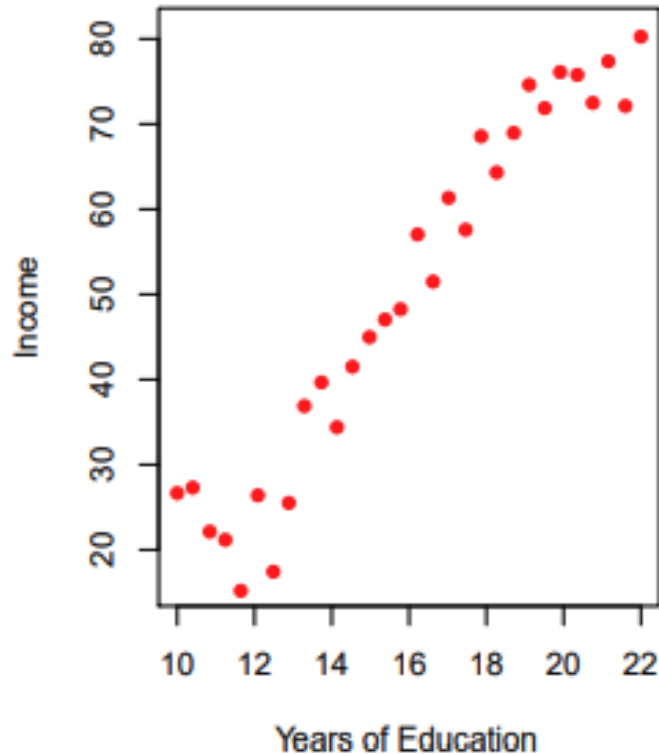


Machine learning

Statistical Learning

1 What is machine learning?

Suppose that we observe a quantitative response Y and p different predictors, $X_1, X_2, \dots, X_p$. We assume that there is some relationship between Y and $X = (X_1, X_2, \dots, X_p)$, which can be written in the very general form:



$$y = f(x) + \epsilon$$

In essence, statistical learning refers to a set of approaches for estimating f

Statistical Learning

① How do we estimate f ?

Parametric models: $f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$

Simpler

Less powerful, can be totally off.

Non-parametric models: do not make explicit assumptions about the functional form of f .

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

$$\min_{\beta} \text{Loss}(Y, \hat{f}(X))$$

Speed

Predictive power

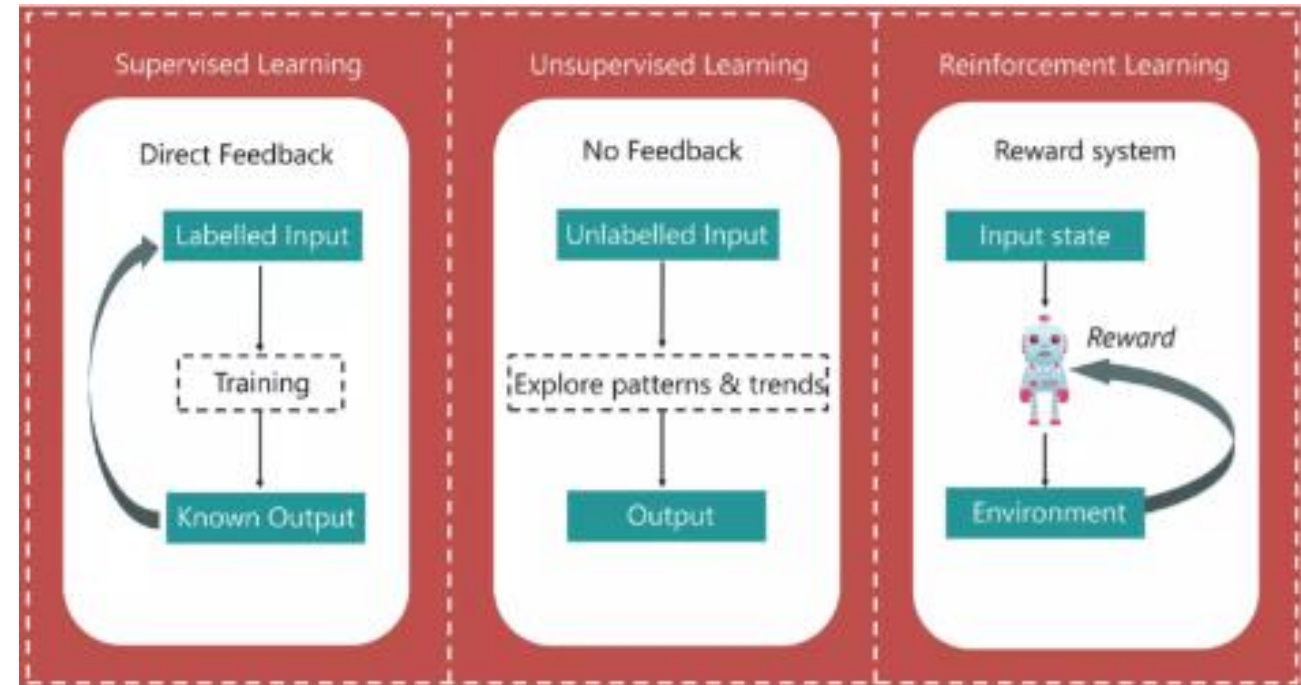
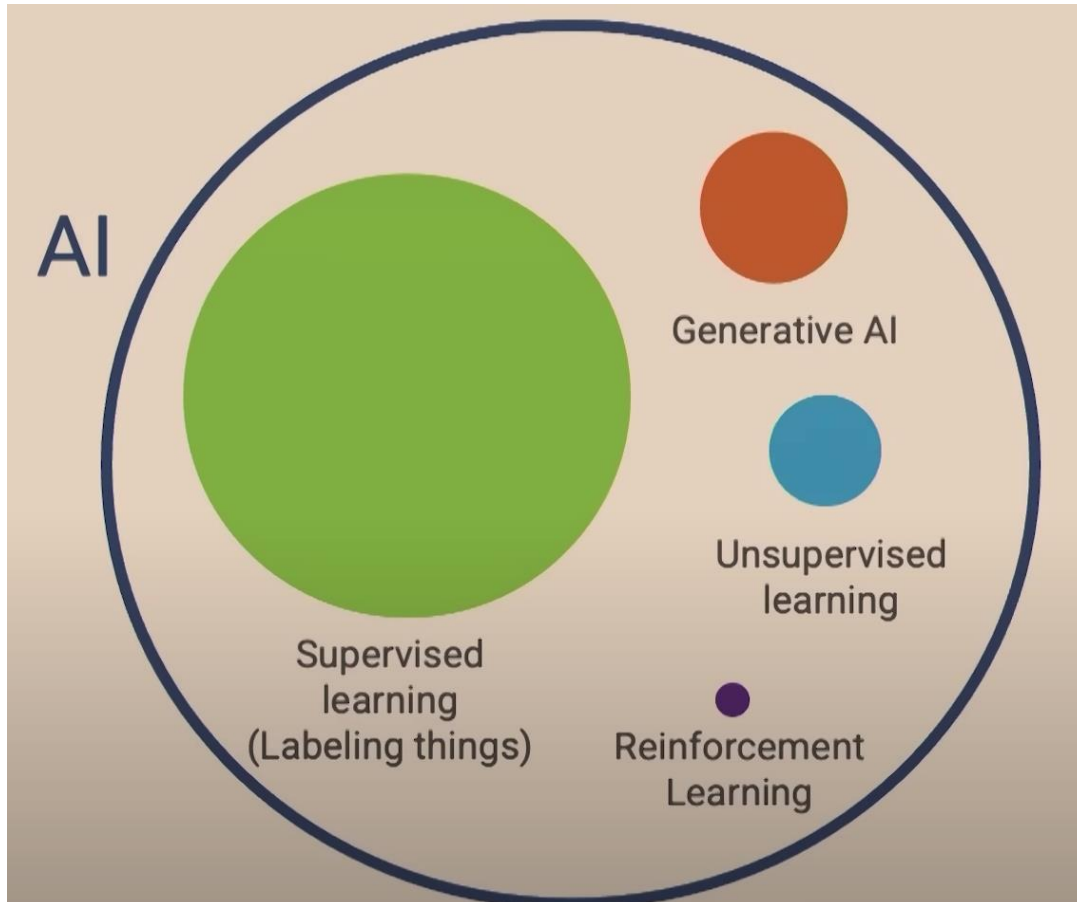
Interpretability

Overfitting

Bias/variance

Statistical Learning

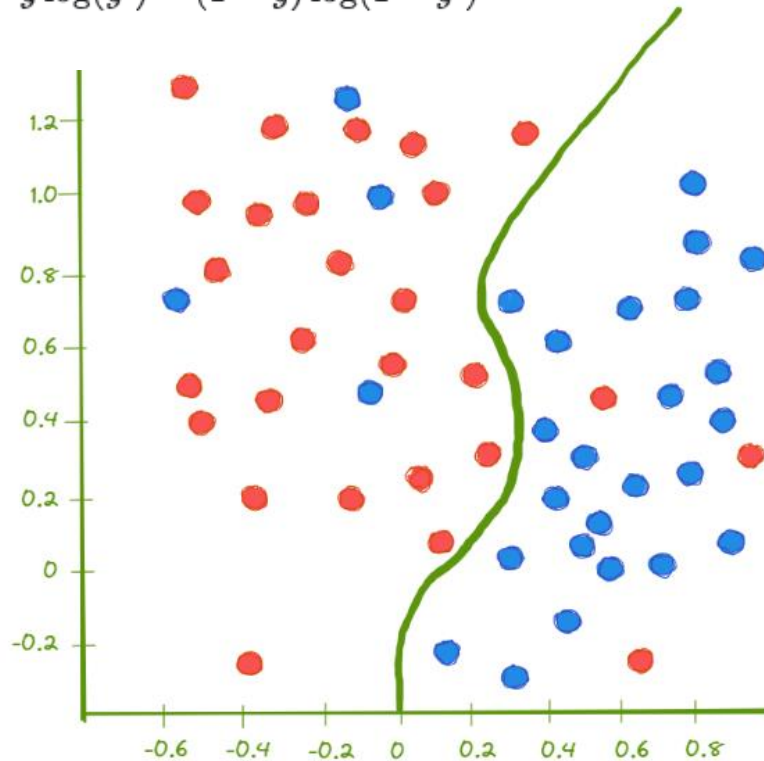
2 Types of learning



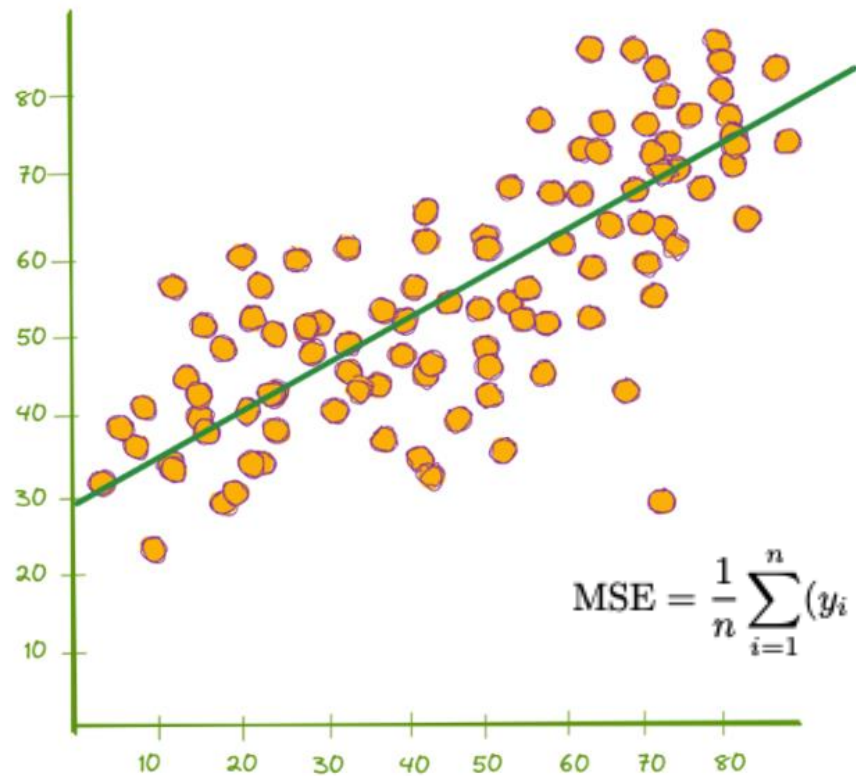
Statistical Learning

2 Supervised Learning

$$\text{Log Loss} = \sum_{(x,y) \in D} -y \log(y') - (1 - y) \log(1 - y')$$



classification

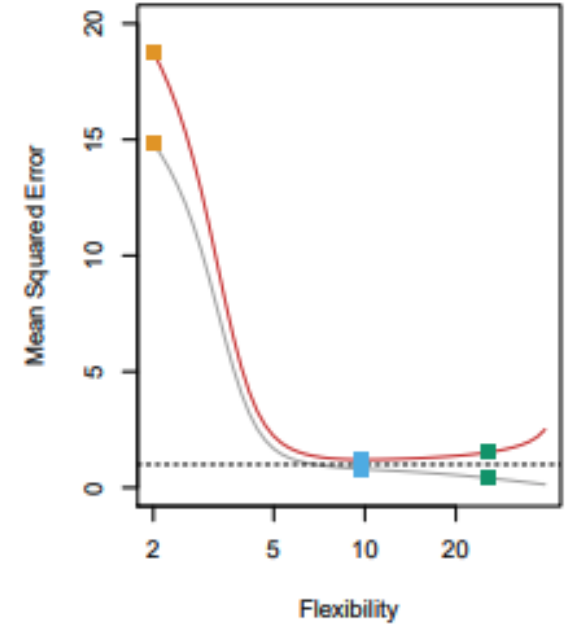
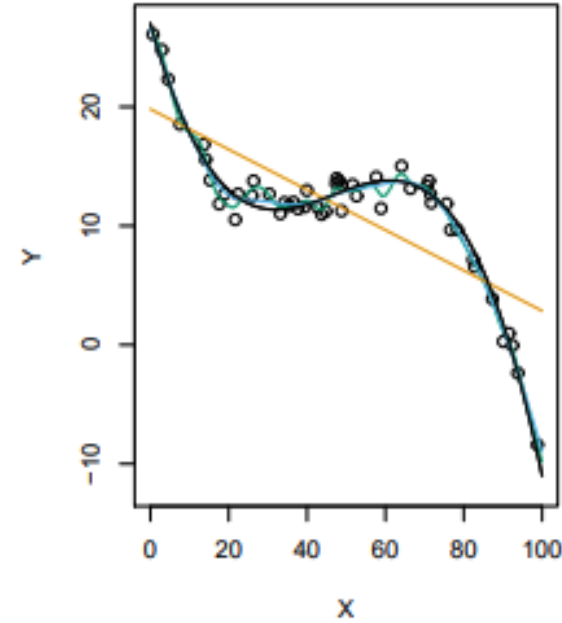
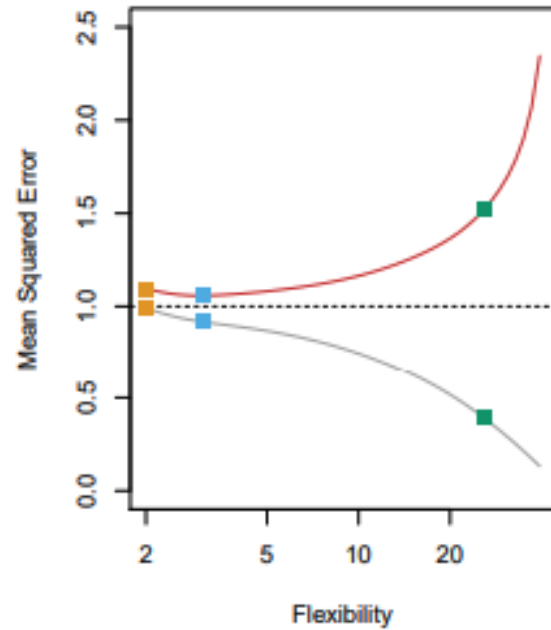
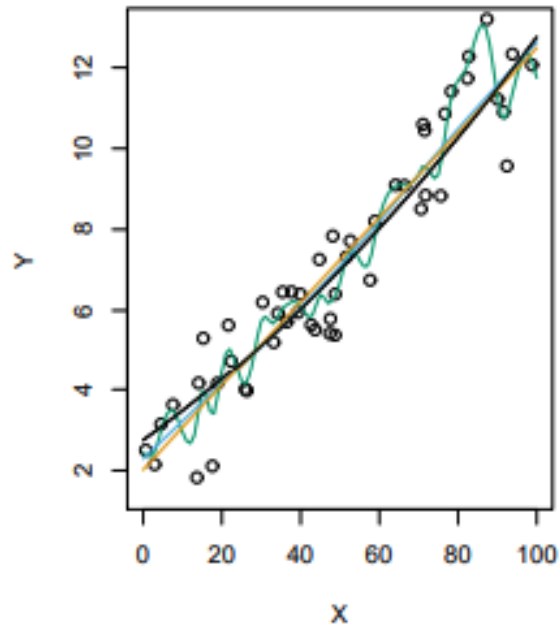


$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2,$$

regression

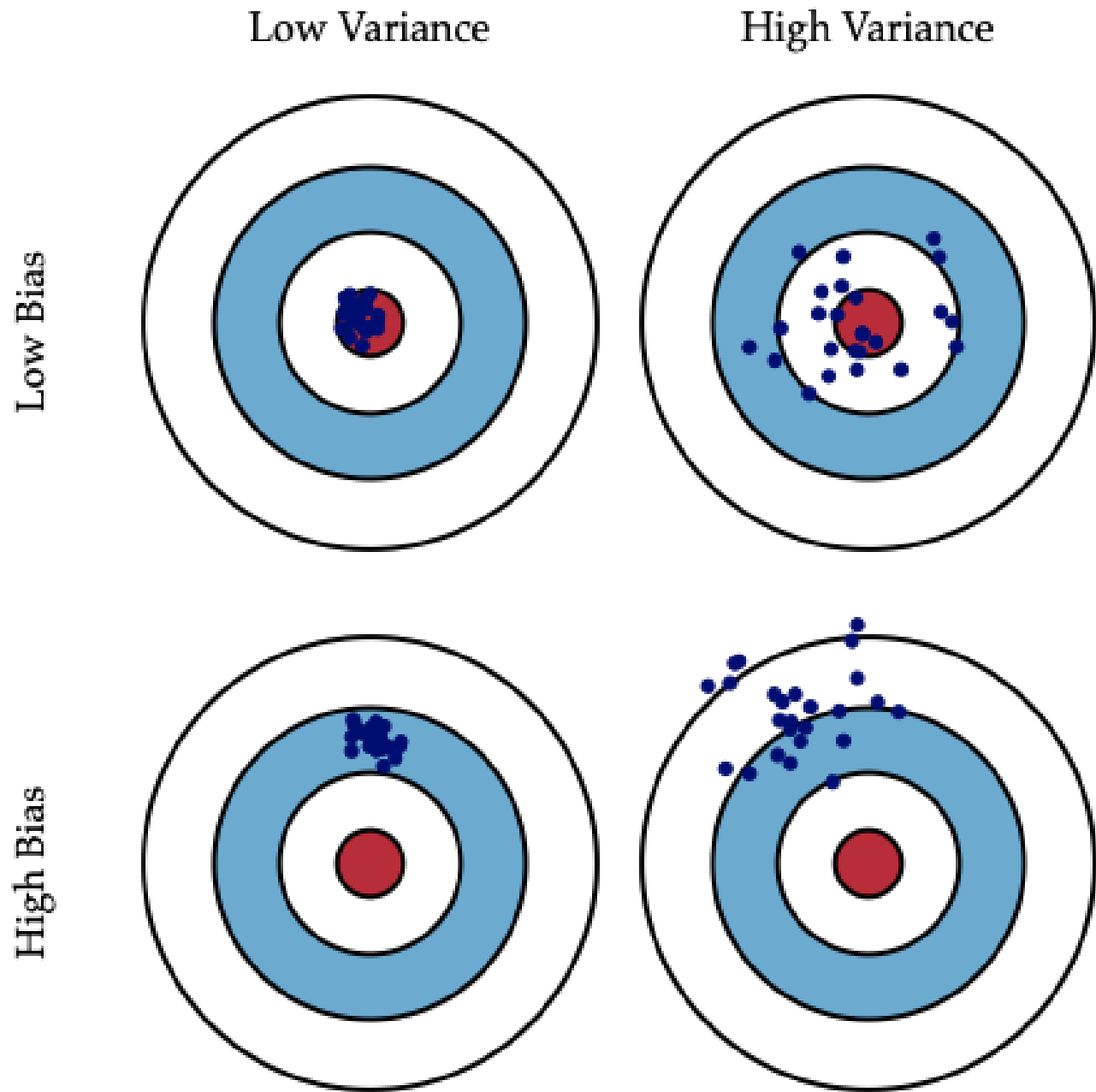
Statistical Learning

3 Bias-variance trade-off



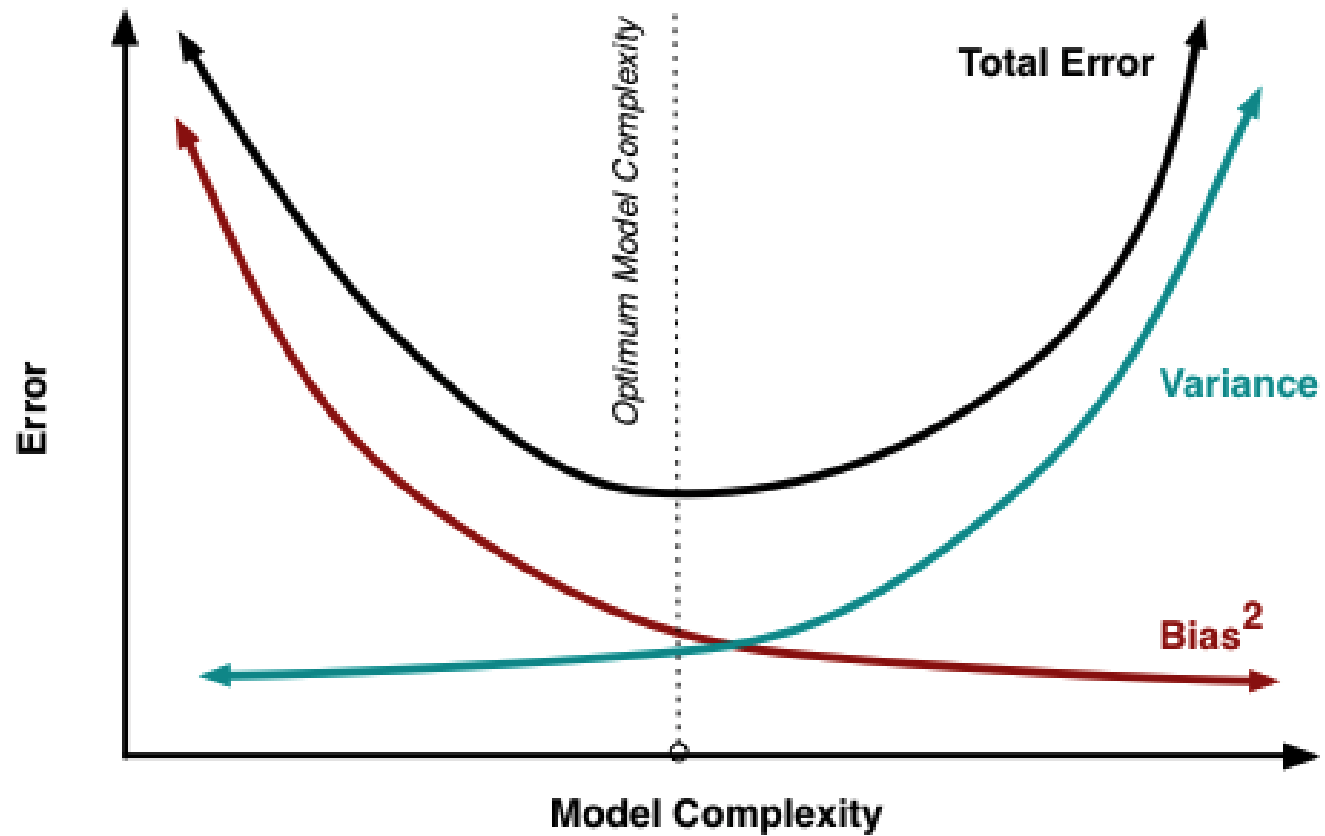
Statistical Learning

3 Bias-variance trade-off



Statistical Learning

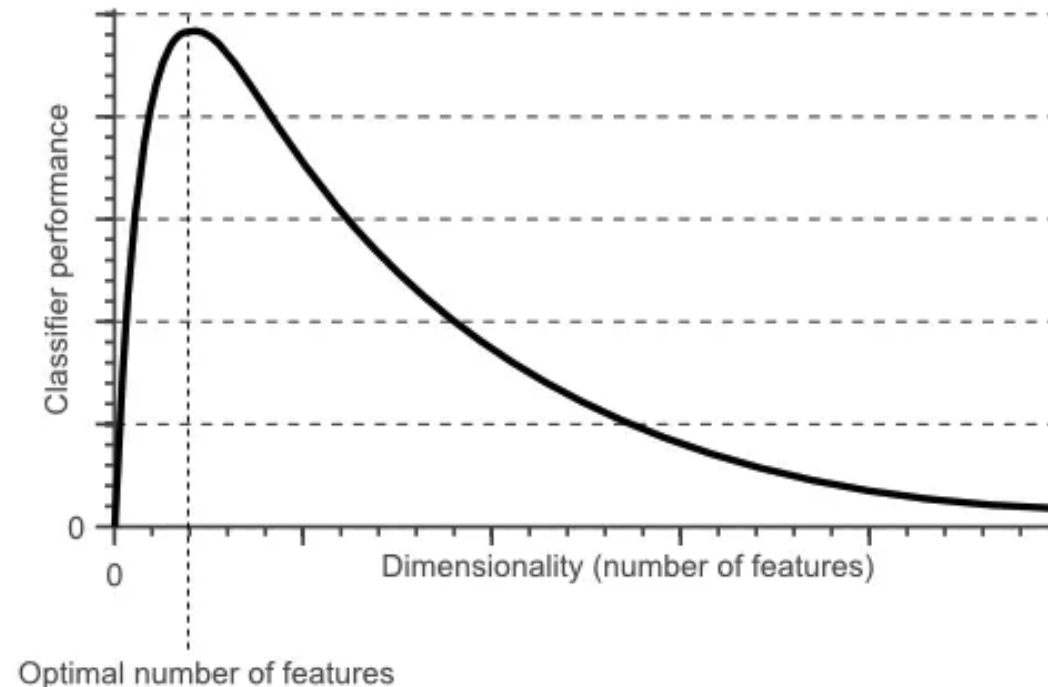
3 Bias-variance trade-off



Statistical Learning

3 Curse of dimensionality

“with a fixed number of training samples, the average (expected) predictive power of a classifier or regressor first increases as the number of dimensions or features used is increased but beyond a certain dimensionality it starts deteriorating instead of improving steadily”



Statistical Learning

4 Linear models

OLS: $\min_w ||Xw - y||_2^2$

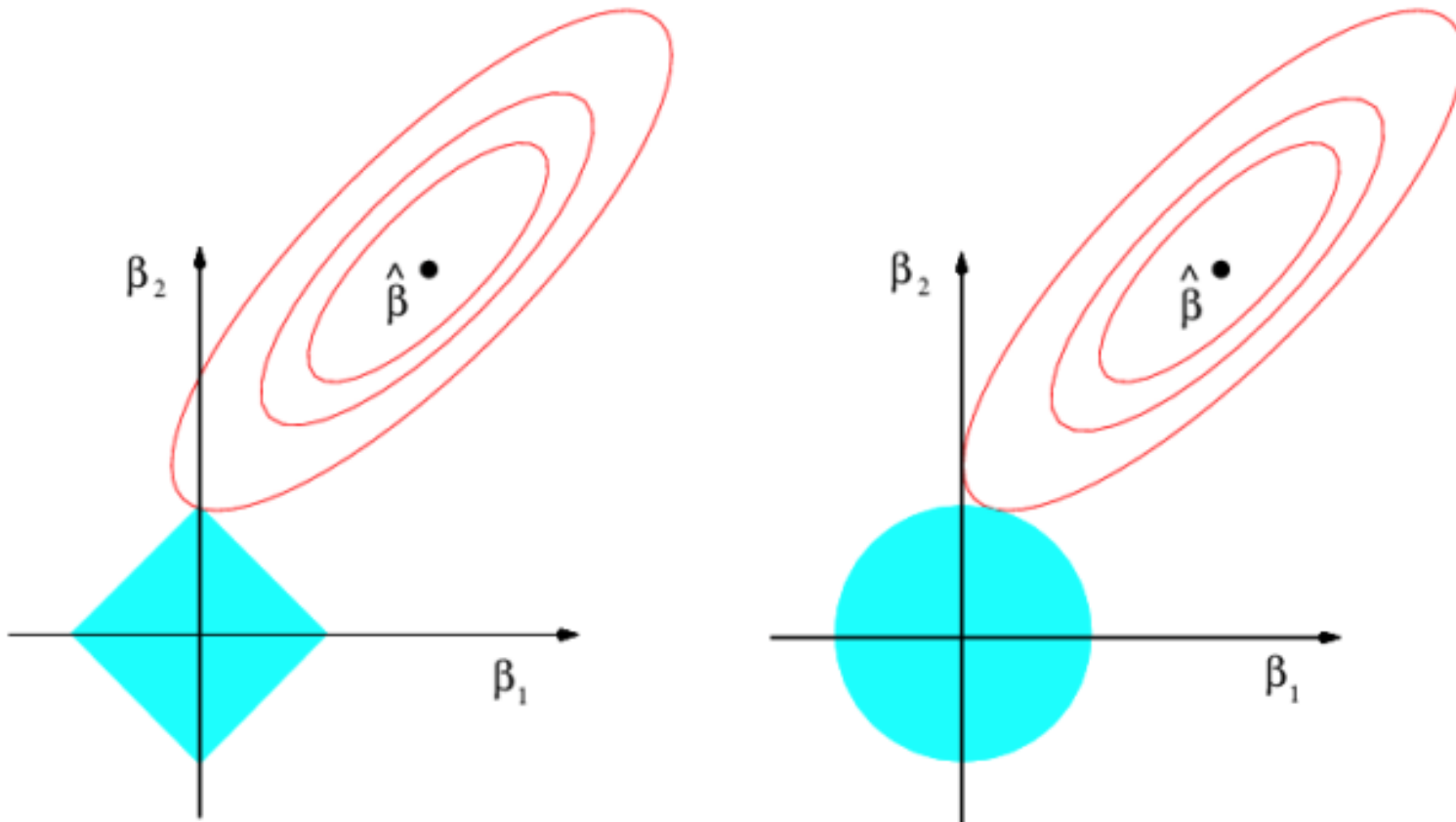
Ridge: $\min_w ||Xw - y||_2^2 + \alpha ||w||_2^2$

Lasso: $\min_w \frac{1}{2n_{\text{samples}}} ||Xw - y||_2^2 + \alpha ||w||_1$

Logistic regression: $\min_w C \sum_{i=1}^n (-y_i \log(\hat{p}(X_i)) - (1 - y_i) \log(1 - \hat{p}(X_i))) + r(w).$

Statistical Learning

4 Linear models. Lasso and Ridge

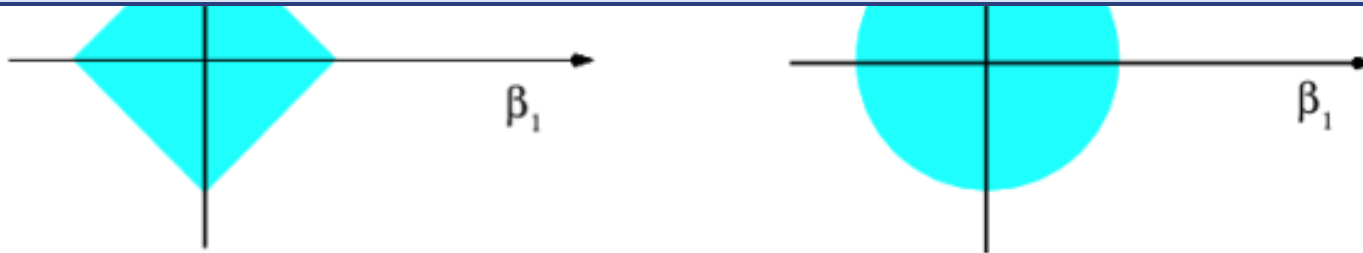


https://hastie.su.domains/ISLP/ISLP_website.pdf -> 6.2.2 The Lasso (interpretación geométrica y bayesiana)

Statistical Learning

4 Linear models. Lasso and Ridge

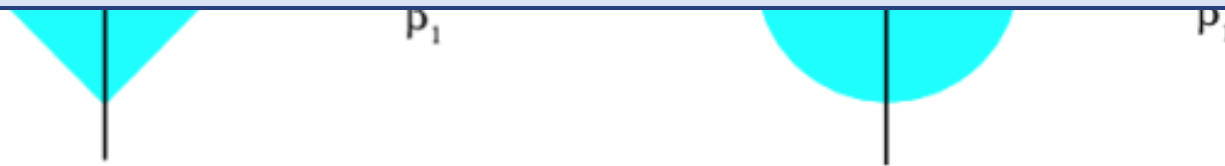
Cómo se relaciona regularisation y el sample size?



Statistical Learning

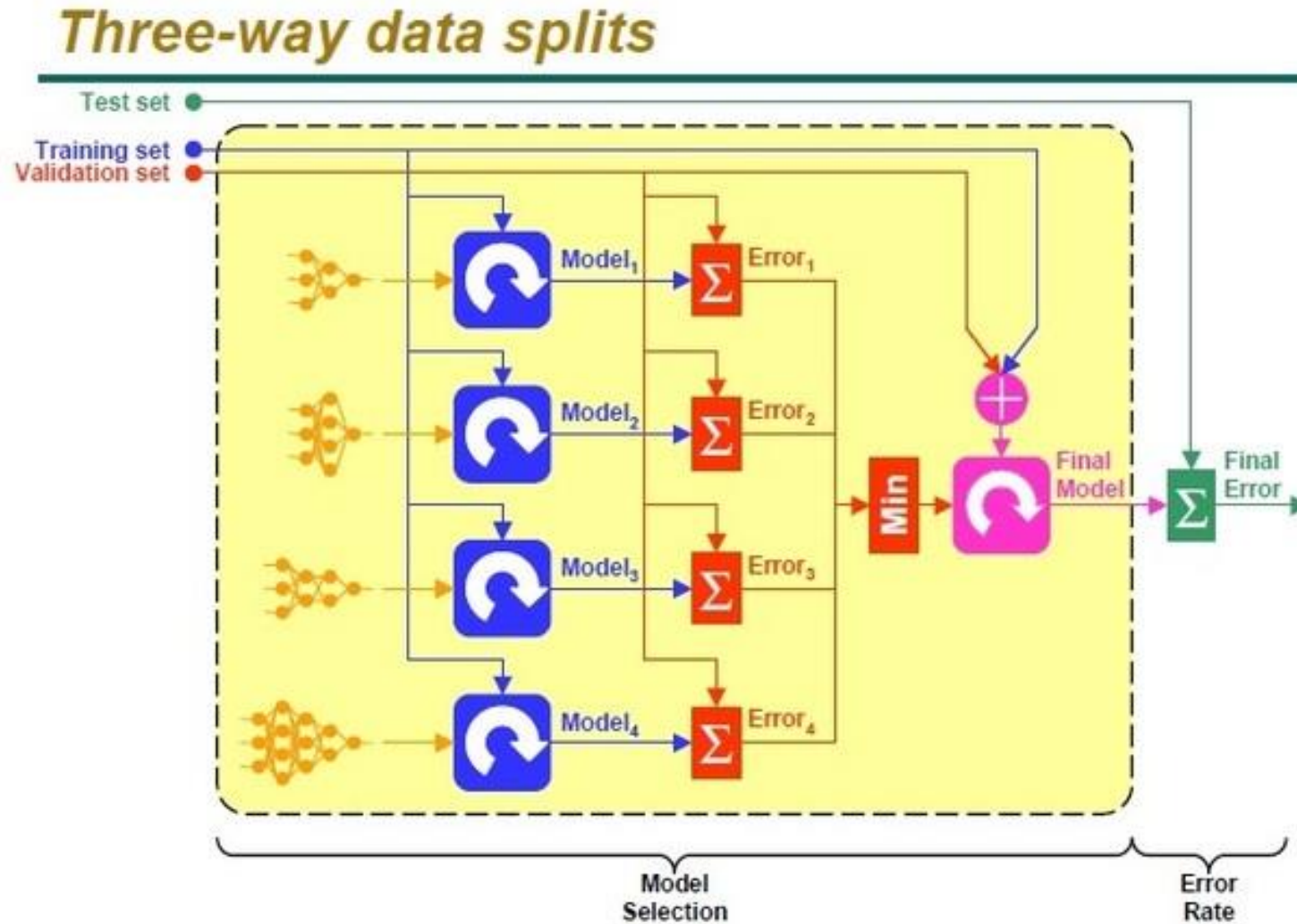
4 Linear models. Lasso and Ridge

Regularization constrains the parameter space of a model that would otherwise overfit the sample. The optimal amount of regularization depends on the model complexity relative to the sample size. Namely, the smaller the sample and/or the complexer the model, the more prone the model is to overfitting.



Statistical Learning

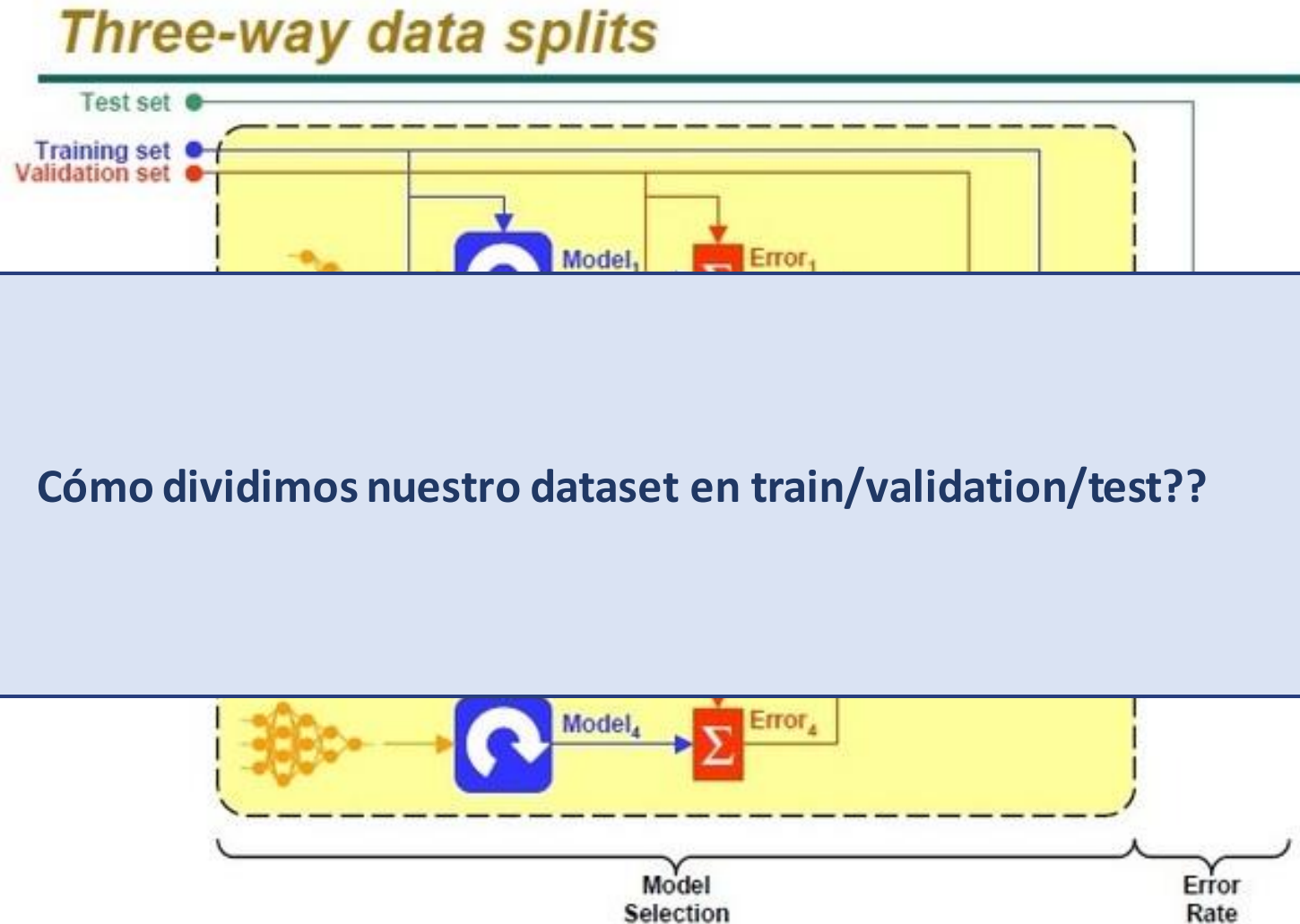
5 3-way split



source: http://research.cs.tamu.edu/prism/lectures/iss/iss_l13.pdf

Statistical Learning

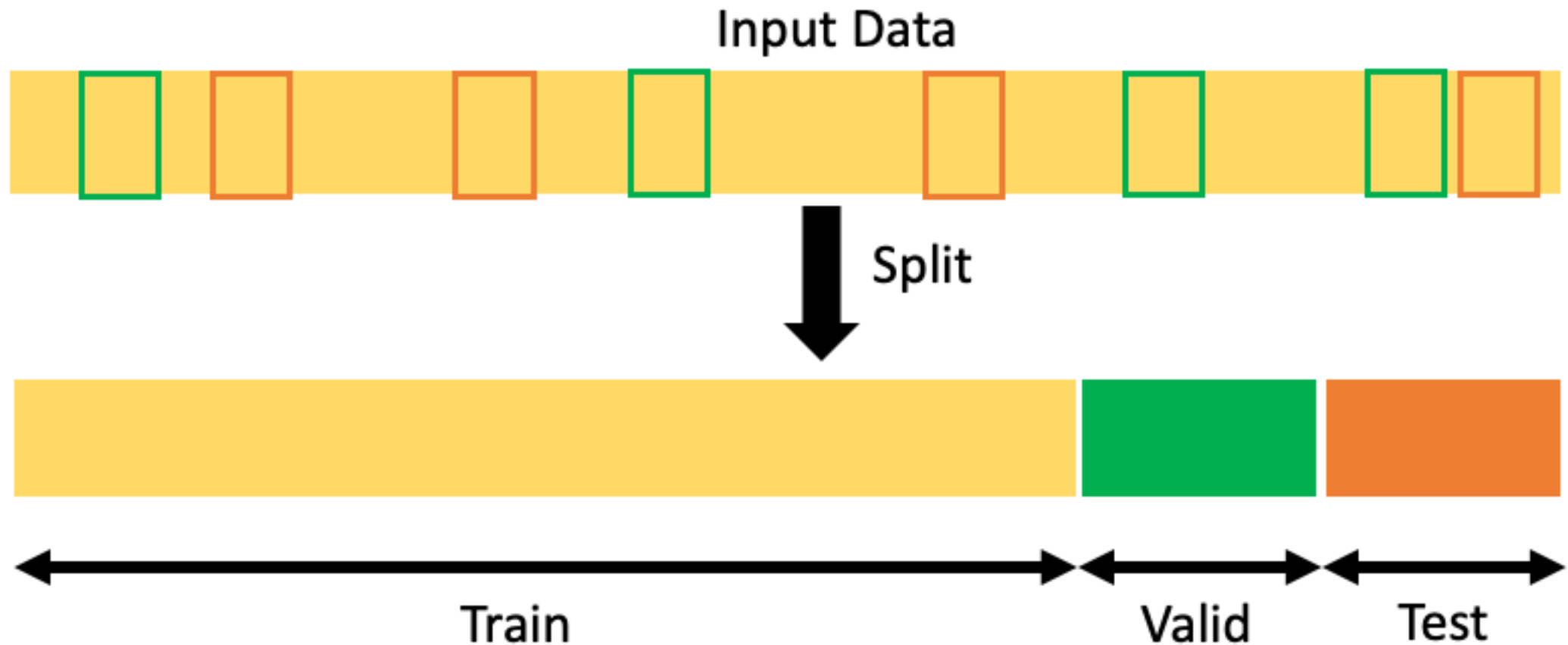
5 3-way split



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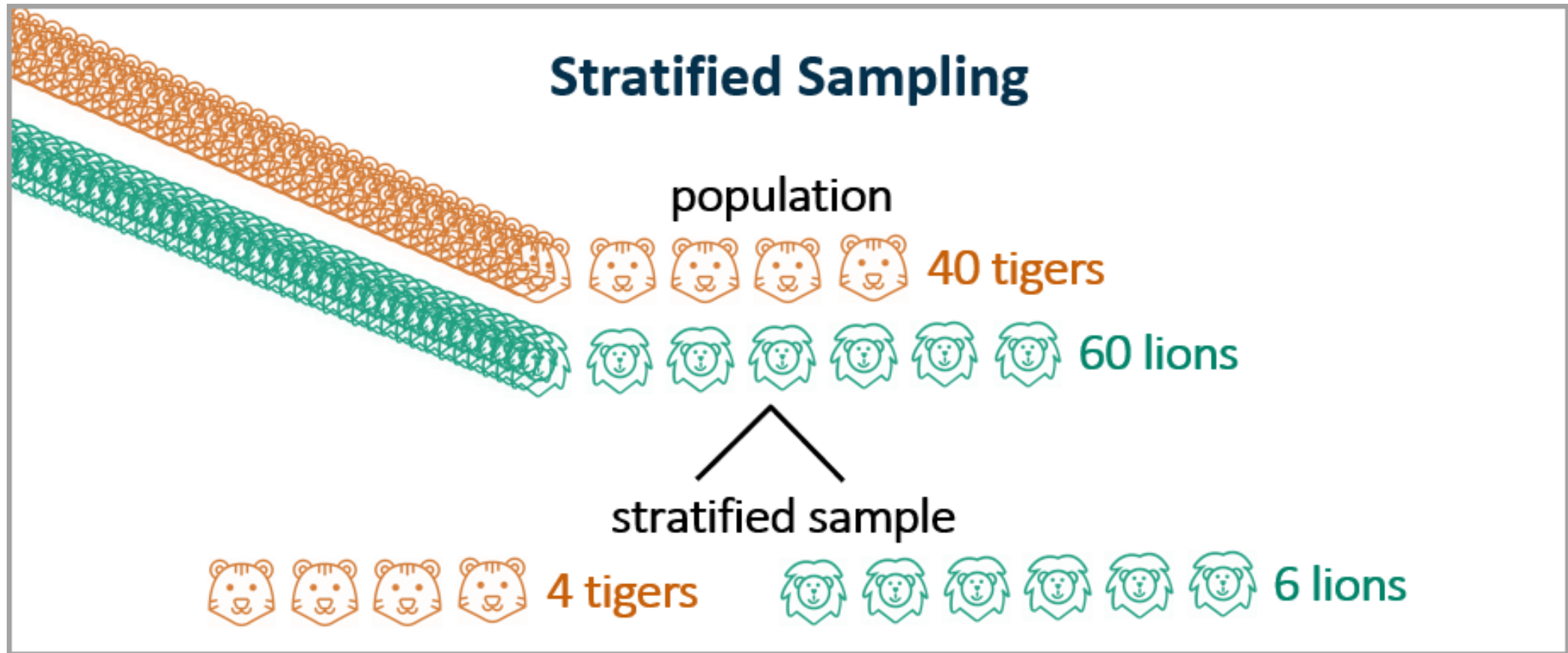
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5 3-way split - Random



Statistical Learning

5 3-way split - Stratified



Statistical Learning

5 3-way split – Advanced considerations

Information leakage: Todo se basa en evitar que en train tengamos información de test.

¿En qué casos pensáis que puede ocurrir utilizando los métodos anteriores?

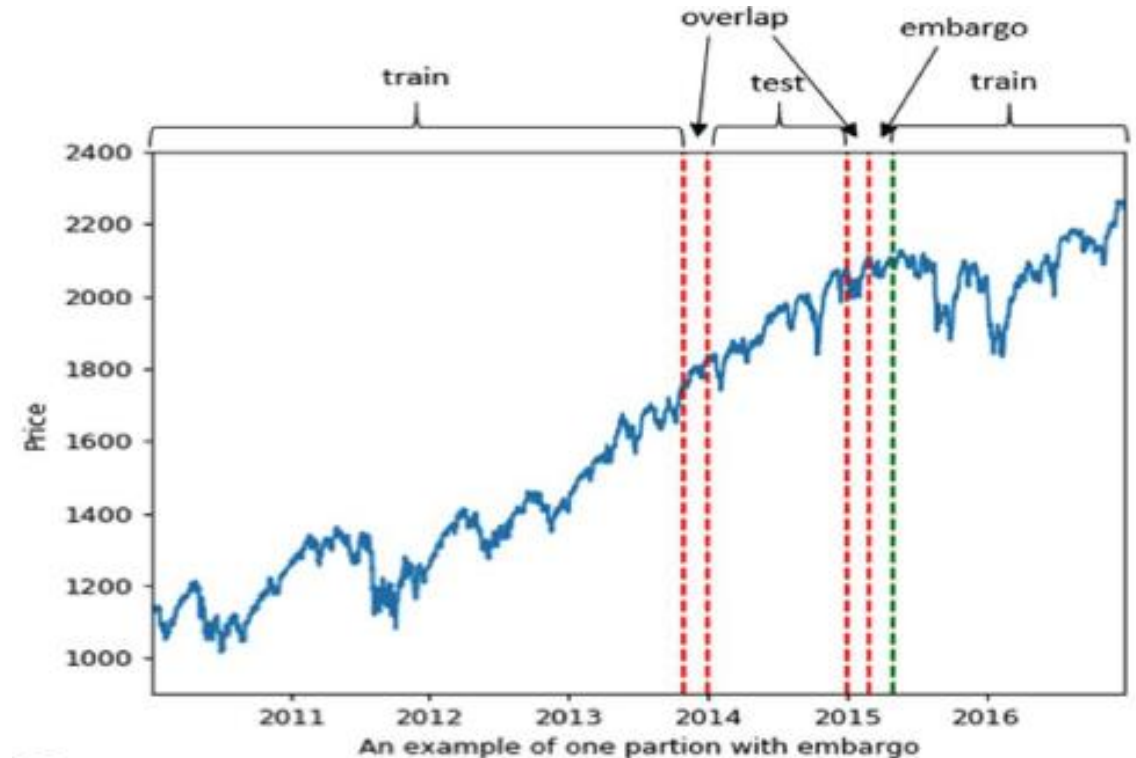
Statistical Learning

5 3-way split – Advanced considerations

Information leakage: Todo se basa en evitar que en train tengamos información de test

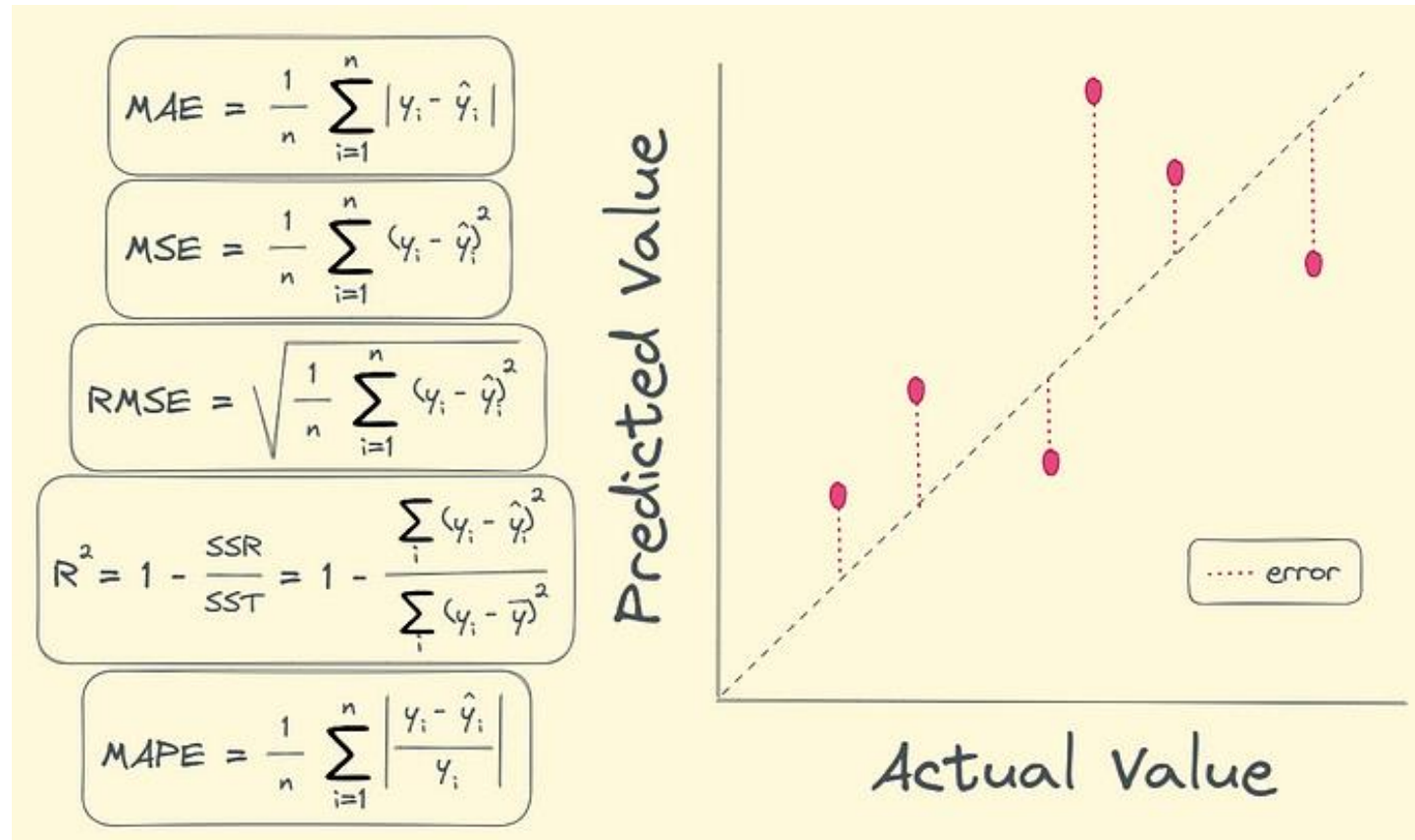
Muestras dependientes:

- Varias muestras de un mismo paciente.
- Múltiples filas correspondientes al mismo usuario.
- Algún tipo de relación temporal.



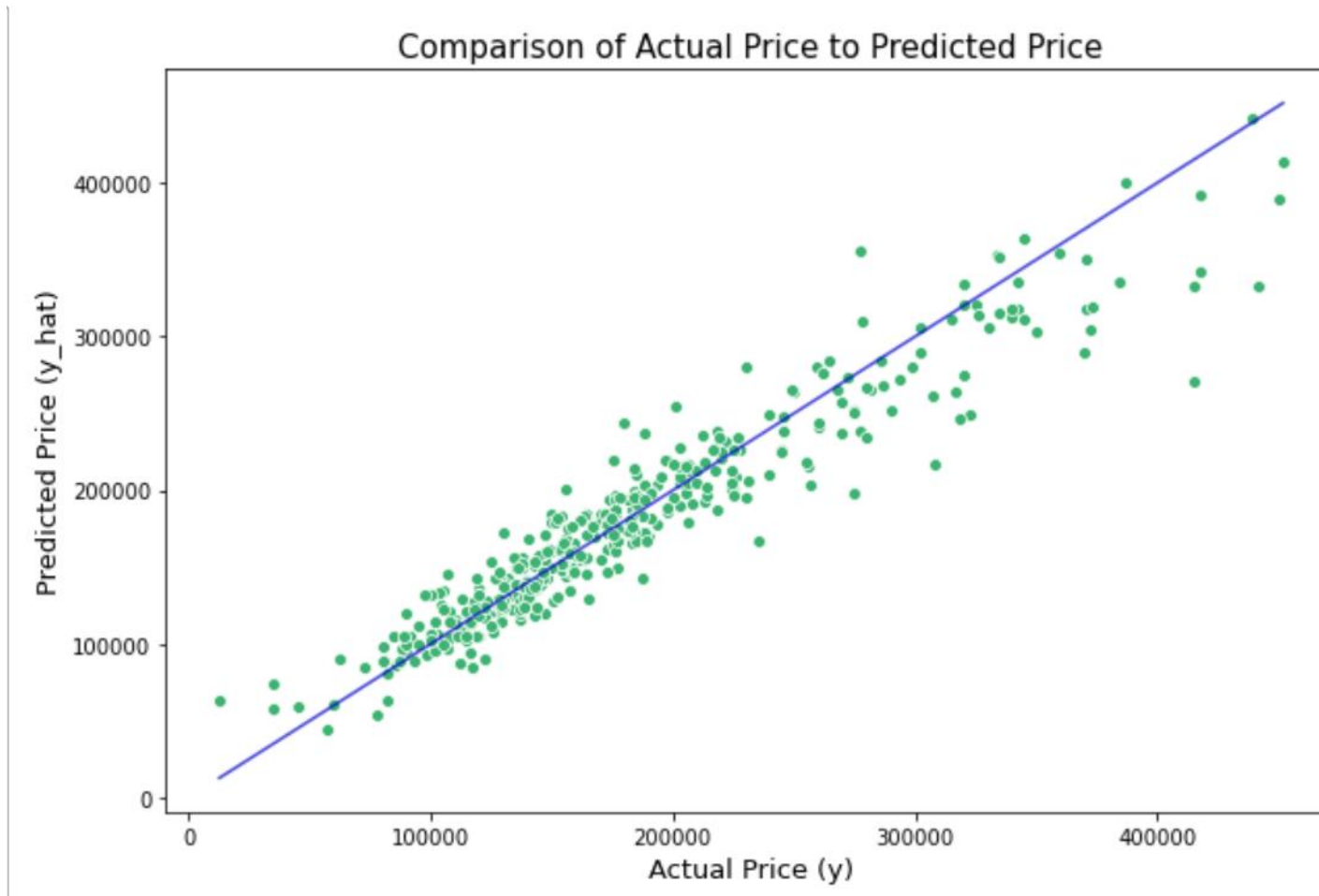
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6 Metrics - Regression



Statistical Learning

6 Metrics - Residuals



Statistical Learning

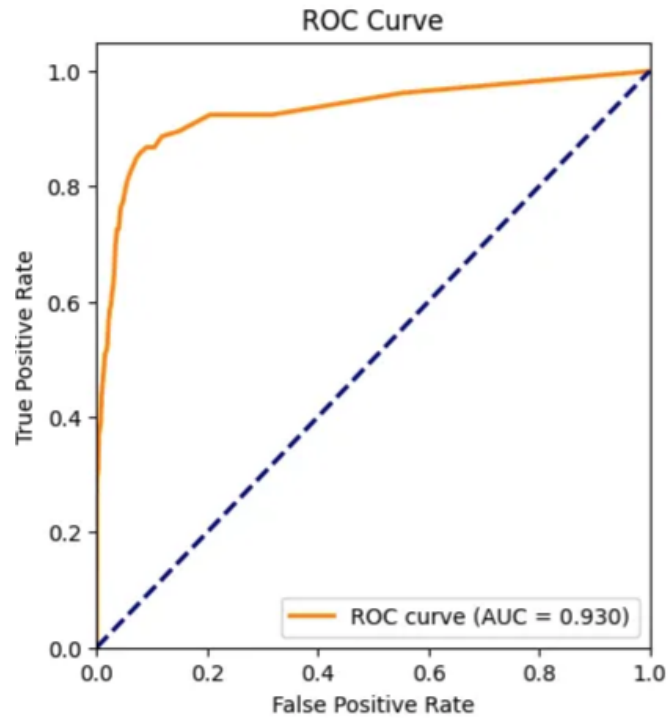
6 Metrics - Classification

		True condition			
Total population		Condition positive	Condition negative	Prevalence $= \frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, $\text{Power} = \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$	
		$F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$			

Statistical Learning

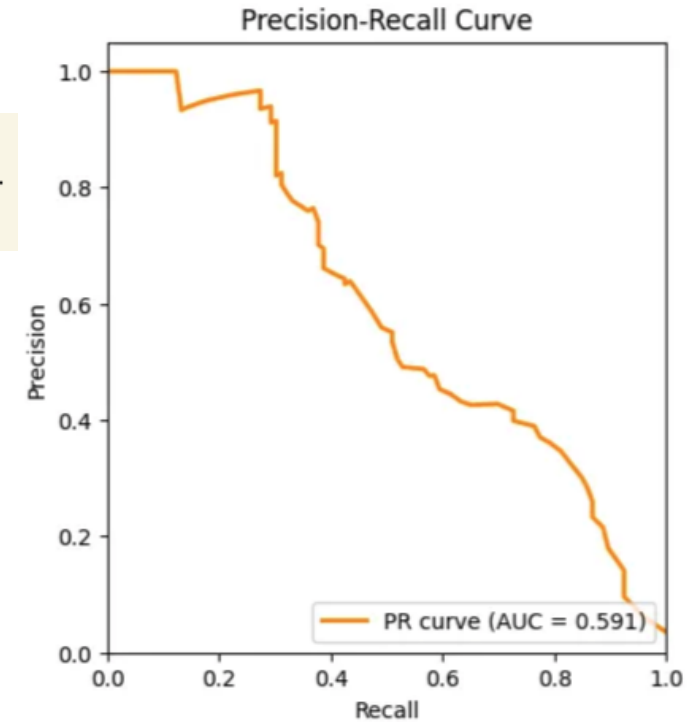
6 Metrics - Classification

$$\frac{TP}{FN + TP} = \frac{TP}{P}$$



$$\frac{FP}{TN + FP} = \frac{FP}{N}$$

$$\frac{TP}{TP + FP}$$

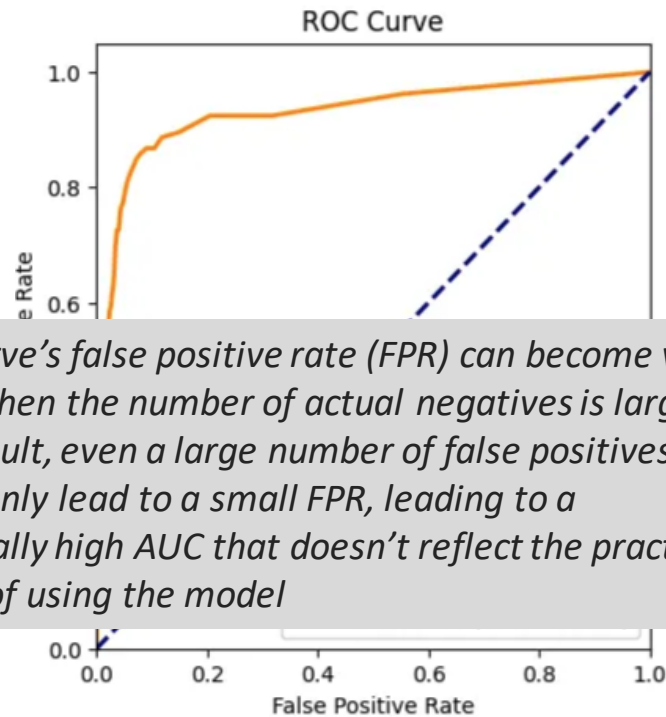


$$\frac{TP}{FN + TP} = \frac{TP}{P}$$

Statistical Learning

6 Metrics - Classification

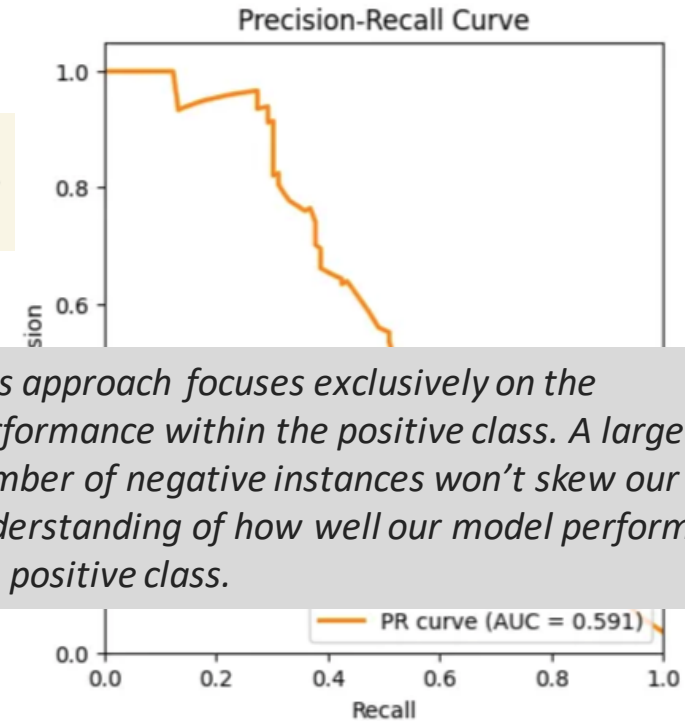
$$\frac{TP}{FN + TP} = \frac{TP}{P}$$



ROC curve's false positive rate (FPR) can become very small when the number of actual negatives is large. As a result, even a large number of false positives would only lead to a small FPR, leading to a potentially high AUC that doesn't reflect the practical reality of using the model

$$\frac{FP}{TN + FP} = \frac{FP}{N}$$

$$\frac{TP}{TP + FP}$$



This approach focuses exclusively on the performance within the positive class. A large number of negative instances won't skew our understanding of how well our model performs on the positive class.

$$\frac{TP}{FN + TP} = \frac{TP}{P}$$

Homework

Q&A